

[13] Selection of Genetic Agents from Random Peptide Aptamer Expression Libraries

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Peptide Aptamers as Genetic Agents

Most cellular processes are regulated by networks of interactions among proteins. Classic "forward" transmission genetic approaches have succeeded in identifying proteins involved in such networks and provide information on the relationships between them. Approaches such as (1) isolation of mutants and mapping the genes responsible for mutant phenotypes, (2) epistasis analysis to determine the order in which gene products might act on each other, (3) dependency analysis to determine the relationship between gene products with respect to the completion of a cellular process,¹ and (4) allele-specific suppression analysis to give clues as to the possible physical interaction between two proteins² provide detailed information about genes and their positions and interactions within networks. Although these approaches are often highly informative, they are often difficult to perform, especially in diploid organisms. Their full application is thus generally limited to a few tractable organisms such as phage, bacteria, yeast, worms, and flies.³

The ability to manipulate individual genes by recombinant DNA approaches allowed the development of "reverse" genetic techniques, in which investigators mutate the function of individual genes and monitor the resulting phenotypes. Reverse genetic approaches that mutate or knock out the function of genes at the DNA level suffer from the following limitations: (1) if the gene under study is essential, then mutations in it will yield nonviable organisms, but often no other information; (2) gene knockouts provide information about the phenotypic consequences of abolishing all the protein interactions in which the knockout gene product participates; (3) gene knockouts typically provide information related only to those phenotypes caused by the initial function of the gene in the development of the organism; and (4) in diploid organisms, as for forward genetic analysis, the observation of recessive phenotypes requires the generation of organisms homozygous for particular mutations, typically in the second generation.

¹ L. M. Hereford and L. H. Hartwell, *J. Mol. Biol.* **84**, 445 (1974).

² P. E. Hartman and J. R. Roth, *Adv. Genet.* **17**, 1 (1973).

³ J. H. Nadeau and P. J. Dunn, *Curr. Opin. Genet. Dev.* **8**, 311 (1998).

More recently, dominant reverse approaches have been devised that affect gene products rather than genes. These dominant reverse approaches rely on generating "mutagenic" agents that inactivate gene products in *trans* without affecting their coding DNA. Examples of dominant genetic agents include small molecule inhibitors,⁴ dominant negative proteins,⁵ injection of antibodies,⁶ antisense RNA,⁷ ribozymes,⁸ and nucleic acid aptamers.⁹ These methods are particularly useful for studying gene function in diploids; however, they too have various weaknesses. Dominant negative proteins and small molecule inhibitors may not exist for all gene products, thus restricting their universality. Antibodies, antisense RNA, ribozymes, and nucleic acid aptamers can in principle be generated to inactivate almost any gene product. Antibodies, however, are not cell permeable and their injection into cells is time consuming and not practicable in all organisms. RNA-based agents are generally not stable in intracellular environments. Although RNA stability can be partly overcome by using RNAs with different chemistry, it remains difficult to predict the sites on the target RNA that are exposed for antisense inhibition. Also, the stability of the protein product of the antisense and ribozyme target gene affects the timing (perdurance) and extent (penetrance) of the mutant phenotype. Moreover, like DNA mutagenesis, mutagenesis at the RNA level results in phenotypes that are caused by the abolishment of all protein interactions in which the target gene product is involved.

A new class of dominant agents has been developed to facilitate the analysis of processes in diploid and genetically intractable organisms. We termed these molecules "peptide aptamers," because of their similarity to nucleic acid aptamers.¹⁰ We define peptide aptamers as antibody-like recognition agents that consist of conformationally constrained peptides displayed on the surface of scaffold proteins, and distinguish such molecules from peptides of variable sequence displayed that are not constrained at both ends by the scaffold. Peptide aptamers are designed to inhibit cellular processes by interacting with proteins and disrupting their biological functions. Combinatorial libraries of peptide aptamers in principle contain aptamers that bind almost any protein target. Peptide aptamers specific for numerous proteins have been isolated by using the yeast two-hybrid system.

⁴ T. J. Mitchison, *Chem. Biol.* **1**, 3 (1994).

⁵ I. Herskowitz, *Nature (London)* **329**, 219 (1987).

⁶ G. J. Gorbisky, R. H. Chen, and A. W. Murray, *J. Cell Biol.* **141**, 1193 (1998).

⁷ A. D. Branch, *Trends Biochem. Sci.* **23**, 45 (1998).

⁸ B. Bramlage, E. Luzi, and F. Eckstein, *Trends Biotechnol.* **16**, 434 (1998).

⁹ M. Thomas, S. Chedin, C. Carles, M. Riva, M. Famulaok, and A. Sentenac, *J. Biol. Chem.* **272**, 27980 (1997).

¹⁰ A. E. Ellington and J. Szostak, *Nature (London)* **346**, 818 (1990).

Using this strategy, we and others have isolated peptide aptamers against Cdk2,¹¹ Ras,¹² human immunodeficiency virus type 1 (HIV-1) Rev,¹³ E2F¹⁴ as well as linear peptides against Rb,¹⁵ and mutagenized PDZ domains against C-terminal peptides.¹⁶ Peptide aptamers recognize their protein targets with K_d values and half-inhibitory concentrations between 10^{-6} and 5×10^{-11} M. The fact that peptide aptamers are typically selected *in vivo* by two-hybrid methods increases the likelihood that the molecules will function *in vivo*. In support of this notion, peptide aptamers function as reverse dominant agents in both mammalian cells^{17,18} and *Drosophila melanogaster*.¹⁹

We and others have also used peptide aptamers as dominant agents for the forward "genetic" analysis of cellular processes.²⁰⁻²² In this approach, combinatorial libraries of peptide aptamers are used as random "mutagens" that interfere with cellular processes. To make this work, the investigator (1) introduces combinatorial expression libraries of peptide aptamers into cells, (2) selects cells that carry peptide aptamers that cause a desired phenotype, and (3) identifies the targets of the peptide aptamers. Target identification is performed by screening the aptamers for interactions against panels of proteins or products of cDNA libraries, using the two-hybrid system. This combination of aptamer-mediated "mutagenesis" with the two-hybrid system to identify aptamer targets provides a general method for characterizing processes in a variety of hitherto genetically intractable organisms.

Here we describe how to obtain and characterize peptide aptamers for both reverse and forward analysis of cellular processes. We describe methods to isolate aptamers against specific proteins with the yeast two-hybrid system as well as methods to identify the targets of aptamers isolated from

¹¹ P. Colas, B. Cohen, T. Jessen, I. Grishina, J. McCoy, and R. Brent, *Nature (London)* **380**, 548 (1996).

¹² C. W. Xu, A. Mendelsohn, and R. Brent, *Proc. Natl. Acad. Sci. U.S.A.* **94**, 12473 (1997).

¹³ B. Cohen, Ph.D. thesis. Harvard University, Boston, Massachusetts, 1998.

¹⁴ E. Fabbriozio, L. Le Cam, J. Polanowski, R. Brent, and C. Sardet, *Oncogene* **18**, 4357 (1999).

¹⁵ M. Yang, Z. Wu, and S. Fields, *Nucleic Acids Res.* **23**, 1152 (1995).

¹⁶ S. Schneider, M. Buchert, O. Georgiev, B. Catimel, M. Halford, S. A. Stackner, T. Baechli, K. Moelling, and C. M. Hovens, *Nature Biotechnol.* **17**, 170 (1999).

¹⁷ B. A. Cohen, P. Colas, and R. Brent, *Proc. Natl. Acad. Sci. U.S.A.* **95**, 14272 (1998).

¹⁸ Xu *et al.*, submitted (2000).

¹⁹ M. G. Kolonin and R. L. Finley, Jr., *Proc. Natl. Acad. Sci. U.S.A.* **95**, 14266 (1998).

²⁰ G. Capronigro, K. R. Abedi, A. P. Hurlburt, A. Maxfield, W. Judd, and A. Kamb, *Proc. Natl. Acad. Sci. U.S.A.* **95**, 7508 (1998).

²¹ C. R. Geyer, A. Colman-Lerner, and R. Brent, *Proc. Natl. Acad. Sci. U.S.A.* **96**, 8567 (1999).

²² T. C. Norman, D. L. Smith, P. K. Sorger, B. L. Drees, S. M. O'Rourke, T. R. Hughes, C. J. Roberts, S. H. Friend, S. Fields, and A. W. Murray, *Science* **285**, 591 (1999).

forward selections. We also illustrate how to use the two-hybrid system to define the specificity of aptamers, to isolate mutant aptamers with enhanced affinity for their targets, and to obtain aptamers specific for allelic variants of proteins. We provide protocols that detail the necessary reagents and techniques to perform the above-described work. While a basic knowledge of yeast and molecular biology techniques is required, descriptions of reagents and techniques not described in this review can be obtained from basic molecular biology manuals.^{23,24}

Design of Intracellular Peptide Aptamers

In Vitro Selection Methods

A number of *in vitro* selection methods have been developed to express randomly encoded peptides (and peptide aptamers) from combinatorial libraries in order to isolate peptides that bind specific targets. These methods range from displaying peptides on the surface of phage,²⁵ yeast,²⁶ DNA-binding proteins,²⁷ polysomes,²⁸ and covalently attached RNA²⁹ to displaying bona fide peptide aptamers on the main flagellar protein of *Escherichia coli*.³⁰ These methods allow isolation of peptides that bind extracellular proteins and proteins immobilized on solid supports. The main advantage of using *in vitro* selection methods to isolate such peptides is the size of the random libraries that can be generated. Currently, libraries containing 10^{10} members are readily obtainable.³¹

The tactics for displaying peptides on the surface of phages, bacteria, or yeast, or in association with the coding RNA, however, all suffer from the following limitations. First, with the exception of the flagellar display of peptide aptamers, the displayed peptides are typically unconstrained

²³ F. M. Ausubel, R. Brent, R. E. Kingston, D. D. Moore, J. G. Seidman, and K. Struhl (eds.), "Current Protocols in Molecular Biology." Greene & Wiley-Interscience, New York, 1992.

²⁴ J. Sambrook, E. F. Fritsch, and T. Maniatis (eds.), "Molecular Cloning: A Laboratory Manual." Cold Spring Harbor Laboratory Press, Cold Spring Harbor, New York, 1989.

²⁵ G. P. Smith and V. A. Petrenko, *Chem. Rev.* **97**, 391 (1997).

²⁶ E. T. Boder and K. D. Wittrup, *Nature Biotechnol.* **15**, 553 (1997).

²⁷ G. M. Gates, W. P. C. Stemmer, R. Kaptein, and P. J. Schatz, *J. Mol. Biol.* **255**, 373 (1996).

²⁸ L. C. Mattheakis, R. R. Bhatt, and W. J. Dower, *Proc. Natl. Acad. Sci. U.S.A.* **91**, 9022 (1994).

²⁹ R. Roberts and J. W. Szostak, *Proc. Natl. Acad. Sci. U.S.A.* (1997).

³⁰ Z. Lu, K. S. Murray, V. Van Cleave, E. R. LaVallie, M. L. Stahl, and J. M. McCoy, *BioTechnology* **13**, 366 (1994).

³¹ J. L. Harrison, S. C. Williams, G. Winter, and A. Nissim, *Methods Enzymol* **267**, 83 (1996).

and of low affinity. Second, none of these selection methods guarantee that the peptide or peptide aptamer will be functional in the cellular environment. For example, peptides may be toxic, they may not bind their target under cellular conditions, and the target may not have the proper conformation or display the same surfaces available for binding. Finally, most *in vitro* selections take place in oxidizing environments, in which disulfide bonds, which are sometimes used to constrain peptides *in vitro*, can form. The cytoplasm and the nucleus are reducing environments, in which disulfide bonds do not typically form.

In Vivo Selection Methods

Peptide aptamers can be directly selected to interact with proteins within cells by the two-hybrid system^{11,32} or by genetically screening for aptamers that cause particular phenotypes.^{20–22} Because these selections take place inside the cell, the aptamers are by definition active and nontoxic. However, when the selection is performed in yeast or in mammalian cells, the number of aptamer-expressing cells that can be easily surveyed ($\sim 10^6$ – 10^7) is typically smaller than the number of peptides that can be surveyed by *in vitro* methods. Libraries of this lower complexity still possess significant biochemical diversity³³ and have been successfully used in both phage display^{34,35} and *in vivo* selections.^{11,15}

Properties of Intracellular Aptamers

The selection of peptide aptamers within cells restricts the types of scaffold proteins and variable region constraints that can be used. Constrained peptide libraries generally produce peptide aptamers with higher binding affinities and greater specificity relative to unconstrained peptide libraries.^{11,36} For *in vitro* selections, the variable region peptide is often constrained either by cyclization through the formation of disulfide bonds³⁷ or by its immobilization on a protein scaffold.³⁸ For selections in the reducing environment of a cell, peptides must either be unconstrained or constrained by means other than disulfide bonds. At least in *E. coli*, unconstrained peptides tend to be unstable. For example, only 5% of the members of a random unconstrained peptide library are able to direct the synthesis

³² Deleted in proof.

³³ M. E. DeGraaf, R. M. Miceli, J. E. Mott, and H. D. Fischer, *Gene* **128**, 13 (1993).

³⁴ M. A. McLafferty, K. A. Kent, R. C. Ladner, and W. Markland, *Gene* **128**, 29 (1993).

³⁵ G. P. Smith, D. A. Schultz, and J. E. Ladbury, *Gene* **128**, 37 (1993).

³⁶ R. Ladner, *Trends Biotechnol.* **13**, 426 (1995).

³⁷ E. Koivunen, B. Wang, and E. Rouslahti, *BioTechnology* **13**, 265 (1995).

³⁸ N. Per-Ake and M. Uhlen, *Curr. Biol.* **7**, 463 (1997).

TABLE I
SCAFFOLDS USED FOR DISPLAYING RANDOM PEPTIDES

Scaffold	Target	K_d	IC ₅₀ (nM)	Ref.
Thioredoxin	Cdk2	30–120 nM	5	11
	Ras	—	100–500	12
	E2F	—	0.05	14
PDZ domain	Peptide	200 nM	—	16
Linear peptide	Rb	10–75 μ M	—	15

of observable protein in *E. coli*.³⁹ Moreover, higher proteolytic stability is expected for peptides that are displayed from a folded structure relative to a linear or unordered peptide. For these reasons, we prefer to use constrained peptide libraries for intracellular selections.

Other desirable properties for aptamer scaffolds include the following: (1) The scaffold should possess good solubility properties, in order to help prevent aggregation of aptamers that contain hydrophobic variable regions; (2) the scaffold should be small, stable, easily purified, and expressed at high levels without toxicity; and (3) the scaffold should be tolerant of modifications such as localization sequences, epitope tags, purification tags, and other protein moieties that allow the aptamer valency to be increased [e.g., aptamers can be dimerized by fusing them to glutathione *S*-transferase (GST)].¹⁸ In the future, we imagine that individual selections will benefit from the use of promiscuous protein scaffolds that can provide additional functional groups for enhanced binding interactions, and through the development of scaffolds carefully designed to display variable regions of different lengths, with different distances between the constrained ends of the variable region, and with different distances between the variable region and the body of the scaffold protein.

Scaffolds Used for Intracellular Aptamers

To date, only a few different scaffolds have been used to display constrained or unconstrained variable regions inside cells. In two-hybrid selections, *E. coli* thioredoxin¹¹ and PDZ domains¹⁶ have been used as scaffolds, and the Gal4 activation domain has been used to display linear peptides.¹⁵ A comparison of the binding constants shows that the aptamers, with their constrained variable regions, bind their targets between 100- and 10,000-fold better than linear peptides (see Table I). For direct phenotypic selec-

³⁹ A. R. Davidson and R. T. Sauer, *Proc. Natl. Acad. Sci. U.S.A.* **91**, 2146 (1994).

⁴⁰ Deleted in proof.

tions, thioredoxin,²¹ green fluorescent protein (GFP),²⁰ and staphylococcal nuclease²² have been used as aptamer scaffolds. Because the dissociation constants for these aptamers were not measured, it is difficult to compare them. However, the fact that the selected aptamers function in two-hybrid experiments suggests that binding constants for all these aptamers are likely to be tighter than the 10^{-6} M threshold for these reporters.

Properties of Thioredoxin Aptamer Proteins

In this review we focus on the properties and uses of *E. coli* thioredoxin (TrxA) as a scaffold protein for constrained variable regions. TrxA was originally used as a scaffold by J. McCoy and co-workers to display peptides on the surface of *E. coli* as a fusion to flagellin.³⁰ Using this approach, TrxA aptamers were obtained against a number of monoclonal antibodies⁴¹ and against protein phosphatase 1.⁴²

TrxA possesses many desirable properties that make it an effective intracellular scaffold protein. TrxA is a small (12-kDa) stable cytoplasmic protein that can be expressed at high levels in cells without toxicity.⁴³ Structural studies have shown that the TrxA active site consists of a short disulfide-constrained peptide loop (–CGPC–).⁴⁴ This loop is tolerant to insertion of peptides⁴³ and thus provides a site to introduce peptides of random sequence. TrxA enhances the solubility of proteins fused to it,⁴³ suggesting that it will enhance the solubility of peptides that would normally aggregate. TrxA can interact with many different protein substrates, so long as those contain disulfide bonds.⁴⁵ The ability of TrxA to interact with many different proteins suggests that it may bear functional groups that make some contacts with the proteins recognized by different aptamer variable regions.

Method 1: Construction of a Thioredoxin Peptide Aptamer Library

Reagents

Vectors

pJM-1: Peptide aptamer vector for use with the interaction trap

pJM-2: Peptide aptamer vector for use in yeast genetic selections

⁴¹ Lu *et al.*, 1995.

⁴² S. Zhao and E. Y. Lee, *J. Biol. Chem.* **7**, 28368 (1997).

⁴³ E. R. LaVallie, E. A. Diblasio, S. Kovacic, K. L. Grant, P. F. Schendel, and J. M. McCoy, *BioTechnology* **11**, 187 (1993).

⁴⁴ S. K. Katti, D. M. LeMaster, and H. Eklund, *J. Mol. Biol.* **212**, 167 (1990).

⁴⁵ B. Wetterauer, M. Veron, M. Miginiac-Maslow, P. Decottignies, and J. P. Jacquot, *Eur. J. Biochem.* **209**, 643 (1992).

pJM-3: Peptide aptamer vector containing a nuclear localization signal for use in yeast genetic selections

See Fig. 1 for a detailed description of the above-described vectors.

Construction of a Random Thioredoxin Peptide Aptamer Library. Thioredoxin peptide aptamer libraries are constructed by inserting an oligonucleotide encoding a random 20-amino acid peptide into an *RsrII* restriction site that occurs naturally in the DNA encoding the TrxA active site loop. This generates a solvent-exposed random 20-amino acid peptide constrained by the active site of TrxA.

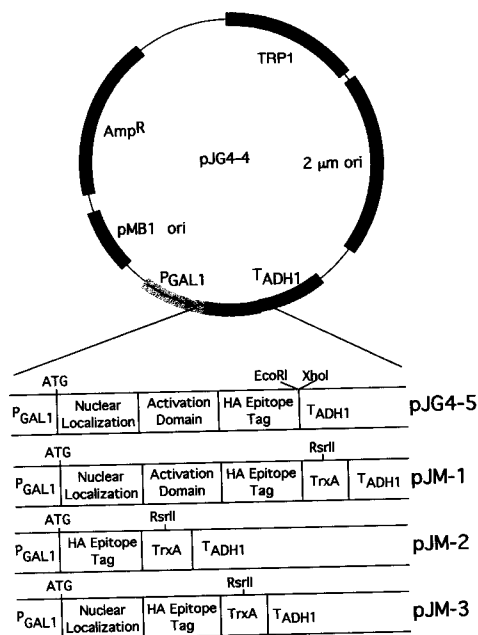


FIG. 1. Expression vectors used for interaction trap and forward genetic selection of peptide aptamers. The yeast-*E. coli* shuttle vectors are derived from pJG4-4.⁴⁷ They contain an *E. coli* origin of replication (pMB1 ori), an ampicillin resistance gene (Amp^R), a *TRP1* marker gene, a yeast 2 μ m origin of replication, and one of the following expression cassettes: pJG4-5, yeast *GAL1* promoter (P_{GAL1}), SV40 nuclear localization signal, B42 activation domain and hemagglutinin epitope tag, *EcoRI* and *XhoI* cloning sites, and yeast *ADH1* transcription terminator (T_{ADH1})⁴⁷; pJM1, yeast *GAL1* promoter (P_{GAL1}), SV40 nuclear localization signal, B42 activation domain and hemagglutinin epitope tag, TrxA, and yeast *ADH1* transcription terminator (T_{ADH1})¹¹; pJM2, yeast *GAL1* promoter (P_{GAL1}) and hemagglutinin epitope tag, TrxA, and yeast *ADH1* transcription terminator (T_{ADH1}) (J. McCoy, personal communication, 1999; and Ref. 21); pJM3, yeast *GAL1* promoter (P_{GAL1}), SV40 nuclear localization signal and hemagglutinin epitope tag, TrxA, and yeast *ADH1* transcription terminator (T_{ADH1}) (J. McCoy, personal communication, 1999; and Ref. 21).

PROTOCOL 1-1: CONSTRUCTION OF RANDOM PEPTIDE APTAMER LIBRARY

1. Choose one of the three TrxA peptide expression vectors (pJM-1, pJM-2, or pJM-3) shown in Fig. 1. pJM-1 is designed for use in the yeast interaction trap two-hybrid system.¹¹ pJM-2 and pJM-3 are, respectively, designed for genetic selections of nuclear-localized and nonlocalized peptide aptamers in yeast.²¹
2. Digest the appropriate pJM vector (Fig. 1) with *RsrII* and dephosphorylate with calf intestine phosphatase.
3. Prepare the following oligonucleotide on an automated DNA synthesizer: 5'-GACTGACTGGTCCG (NNG/C)₂₀GGTCCTCAGTCAGTCAG, where N represents G, C, A, or T.
4. Add 200 μ g (a 10-fold excess) of the primer 5'-CTGACTGACTGAGGACC to 100 μ g of the above-described random oligonucleotide to give a total volume of 890 μ l in 1 $\times$ Klenow polymerase buffer. Anneal the primer by heating the sample to 94° for 5 min and slowly cool to room temperature.
5. Add 90 μ l of a 5 mM mixture of the four dNTPs and 20 μ l of Klenow polymerase (5 U/ μ l) and incubate the reaction for 3 hr at 37°.
6. Phenol–chloroform extract the mixture. Follow this with a chloroform extraction.
7. Ethanol precipitate and wash the pellet with 80% (v/v) ethanol. Dissolve the pellet in 0.9 ml of H₂O.
8. Add 0.1 ml of *AvaII* buffer and digest the DNA with 1000 units of *AvaII* for 4 hr. *Optional*: Digest the DNA from step 7 with 1000 units of *BfaI* overnight. *BfaI* cleaves DNA containing CTAG and eliminates TAG stop codons following C.
9. Phenol–chloroform extract the mixture, followed by a chloroform extraction.
10. Ethanol precipitate the restriction digest and dissolve the DNA in 10 mM Tris (pH 8) and a 1/5 volume of nondenaturing loading buffer.
11. Purify the DNA on a 10% nondenaturing polyacrylamide gel. Expose the DNA by UV shadowing. Cut out the DNA band and elute in 10 mM Tris, pH 8, overnight. Ethanol precipitate the supernatant and dissolve in 10 mM Tris, pH 8.
12. Ligate 8 μ g of the random sequence DNA insert into 12 μ g of the *RsrII*-cut dephosphorylated pJM vector from step 1 with 80,000 units [New England BioLab (Beverly, MA) units] of ligase in a 1-ml reaction.
13. Purify the DNA with a QIAquick gel extraction kit from Qiagen (Chatsworth, CA) and elute with 30 μ l of ultrapure H₂O.
14. Add 30 μ l of DNA to 350 μ l of competent *E. coli* (e.g., MC1061; Bio-Rad, Hercules, CA) in 0.2-cm gap electroporation cuvettes. Electroporate,